No.	Signature	Interpretation	Attack Labels	Individual Precision
1	Average packet length: ⟨42.0, 42.0⟩ ∧ Maximum packet length: ⟨42.0, 42.0⟩ ∧ Packet length radius: ⟨0.0, 0.0⟩ ∧ Total sum of packet lengths: ⟨441.0, 441.0⟩ ∧ Number of packets: ⟨9.5, 9.5⟩ ∧ Traffic weight: ⟨141.55, 141.55⟩ ∧ HTTPS traffic: ⟨0, 0⟩	Uniform 42-byte non-HTTPS traffic. The packet window has an average and maximum length of 42 bytes with zero length radius, indicating highly uniform small packets. This pattern is compatible with ICMP or UDP flooding but lacks protocol and rate conditions that distinguish them, while Number and Weight describe window statistics rather than exact volume. Operators can verify it using the IP protocol, sustained packet rate, target concentration, source diversity, and recurrence across windows.	ddos-icmp_-flood, dos-udp_-flood	100%
2	HTTP traffic: ⟨0, 0⟩ ∧ Minimum packet length: ⟨42.0, 42.0⟩ ∧ UDP traffic: ⟨0, 0⟩ ∧ URG flag count: ⟨0.0, 0.0⟩ ∧ Number of packets: ⟨9.5, 9.5⟩ ∧ Traffic weight: ⟨141.55, 141.55⟩ ∧ HTTPS traffic: ⟨0, 0⟩	Non-web, non-UDP traffic with a 42-byte minimum packet. The window is not classified as HTTP, HTTPS, or UDP and contains no URG flags, while Number and Weight do not by themselves establish repeated packets or flooding volume. Although this pattern matches ICMP-flood records, it lacks explicit ICMP and rate conditions. Operators should check the IP protocol, ICMP type, packet rate, echo pattern, target concentration, and source distribution.	ddos-icmp_-flood	100%
3	Protocol type: ⟨6.0, 6.0⟩ ∧ RST flag count: ⟨0, 0⟩ ∧ SYN flag count: ⟨0.0, 0.0⟩ ∧ UDP traffic: ⟨0, 0⟩ ∧ Number of packets: ⟨9.5, 9.5⟩ ∧ Traffic weight: ⟨141.55, 141.55⟩ ∧ HTTPS traffic: ⟨0, 0⟩	TCP traffic without observed SYN or RST flags. The window represents non-HTTPS TCP traffic outside the observed connection-opening and reset phases, while Number and Weight do not establish high traffic volume. This broad pattern may occur in ACK or PSH-ACK floods and normal established TCP traffic, but it does not alone explain the other listed attacks. Operators should inspect ACK, PSH, and FIN flags, fragmentation, HTTP fields, flow duration, packet rate, destination fan-out, and network-layer protocol.	ddos-ack_-fragmentation, ddos-pshack_flood, ddos-rstfinflood, ddos-slowloris, ddos-tcp_-flood, dos-http_-flood, dos-syn_-flood, dos-tcp_flood, mirai-greip_-flood, vulnerabilityscan	99%

Table 11: Three representative signatures from `CICIoT2023`. Those signatures are used to detect denial-of-service attacks, which accounts for most of the attacks in the dataset.